

1. Define flow duration curve of a river. Describe the role of flow discharge curve of a river in the hydraulic design of headworks and powerhouse structures.

Answer:

A Flow Duration Curve (FDC) is a plot showing the percentage of time that a given discharge is equaled or exceeded. It is widely used to determine the design discharge and hence the installed capacity of a Run-of-River hydropower plant.

Hydropower output is given by:

$$P = \rho g Q H \eta$$

where:

$$P = \text{Power (W)}$$

$$\rho = \text{Density of water (1000 kg/m}^3\text{)}$$

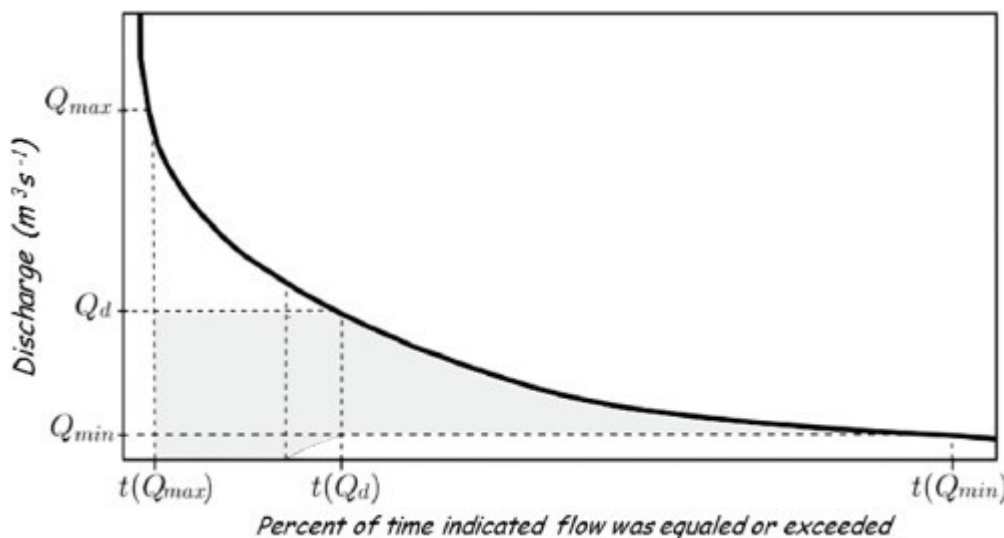
$$g = \text{Acceleration due to gravity (9.81 m/s}^2\text{)}$$

$$Q = \text{Design discharge (m}^3\text{/s)}$$

$$H = \text{Net head (m)}$$

$$\eta = \text{Overall efficiency}$$

$$P = \frac{9.81 Q H \eta}{1}$$



Role of flow discharge curve of river in hydraulic design of headworks:

- It helps to determine the discharge to be diverted into the canal or power intake.
- It also ensures optimum utilization of available river flow without oversizing structures.
- Flow duration curve is also used to determine the dimensions of intake openings, gates, and approach channels.
- The dependable discharge obtained from the FDC is used to design the carrying capacity of irrigation canals.
- It indicates periods of high discharge when sediment concentration is generally higher.
- Flow duration curve also helps in designing desanders, settling basins, and flushing arrangements.

Role of flow discharge curve of river in hydraulic design of powerhouse structures:

- It prevents underutilization or excessive investment.

- ii. The curve also determines the optimum discharge for turbine operation and ensures maximum annual energy production.
- iii. The dependable discharge from the FDC is used to estimate firm power generation during dry seasons.
- iv. Flows above the design discharge indicate the amount of surplus or secondary energy that can be generated during wet seasons.
- v. Flow variation information is used to size tailrace channels and draft tubes to ensure efficient turbine performance.
- vi. The curve also helps balance the cost of turbines, generators, penstocks, and civil works against expected energy generation.

The FDC provides dependable discharge, which is the flow equaled or exceeded for a specified percentage of time.

- **50% dependable flow:** Average flow available half of the time.
- **75% dependable flow:** Used for medium reliability designs.
- **90–95% dependable flow:** Used for firm power estimation and reliable water supply.

2. Define probable maximum flood (PMF). Explain its importance in design hydraulic structures.

Answer:

- Probable Maximum Flood is the maximum flood that occurs under the worst hydrological and meteorological conditions.
- It is the extreme flood that is physically possible in the region as a result of severe most combinations.
- PMF is obtained from the study of all the storms that have occurred over a region and maximizing them for the most critical atmospheric condition.
- PMF is used to ensure the safety and stability of hydraulic structures whose failure could result in catastrophic consequences.
- The importance of probable maximum flood in the design of hydraulic structures are listed in the points below:
 - i. PMF is used for the design of spillway to safely pass the PMF without overtopping them.
 - ii. It prevents the structural failure of the hydraulic structures due to excessive flood discharge.
 - iii. PMF is also essential for the design of large storage dams, embankments dams and ensure its stability under the extreme flood conditions.
 - iv. It also prevents catastrophic downstream flooding caused by dam failure.
 - v. It reduces the risk to human lives, infrastructures and agricultural land.
 - vi. PMF is also used to determine the capacity of spillways, flood control structures, energy dissipators etc.
 - vii. The Probable Maximum Flood also ensures reliable performance of structures during extreme flood events.
 - viii. Although designing for PMF increases initial construction cost, it greatly reduces the risk of catastrophic failure and associated economic losses.

3. What are the basic equations for steady flow? What are the major causes for head loss in pipelines? Elaborate pipe friction head loss in pipelines.

Answer:

In a steady flow, the flow characteristics such as velocity, pressure, discharge, etc. at any point do not change with time. The basic equations governing steady flow are:

- i. Continuity equation (Conservation of mass)

$$Q = AV = \text{constant}$$

or

$$A_1 V_1 = A_2 V_2$$

where :

$$Q = \text{Discharge} (m^3 / s)$$

$$A = \text{Cross - sectional area} (m^2)$$

$$V = \text{Flow velocity} (m / s)$$

ii. Bernoulli's Equation:

Bernoulli's equation states that the total energy of flowing water remains constant along a streamline, neglecting energy losses.

$$\frac{P}{\gamma} + \frac{V^2}{2g} + Z = \text{constant}$$

Including head loss:

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2 + h_f$$

where:

$$P / \gamma = \text{Pressure head}$$

$$V^2 / 2g = \text{Velocity head}$$

$$Z = \text{Elevation head}$$

$$h_f = \text{Head loss due to friction}$$

Major causes of head loss in pipes:

Head loss is the reduction in total energy of flowing water due to resistance.

i. Major head loss:

Friction between flowing water and the inner surface of the pipe.

ii. Minor head loss:

- Entrance loss
- Exit loss
- Sudden expansion
- Sudden contraction
- Bends and elbows
- Valves and fittings
- Tees and junctions
- Obstructions and meters

Pipe friction head loss in pipelines:

Pipe friction head loss is the major energy loss occurring due to friction between the flowing fluid and the pipe wall. It depends on pipe length, diameter, roughness, velocity, and flow conditions.

The most commonly used equation is the Darcy–Weisbach equation:

$$h_f = f \frac{L}{D} \frac{V^2}{2g}$$

where:

$h_f = \text{Friction head loss (m)}$

$f = \text{Darcy friction factor}$

$L = \text{Length of pipe (m)}$

$D = \text{Diameter of pipe (m)}$

$V = \text{Mean velocity (m/s)}$

$g = \text{Acceleration due to gravity (9.81 m/s}^2\text{)}$

Factors affecting friction head loss:

- i. Length of pipe (L):
Longer the length of pipe, greater the head loss.
- ii. Velocity of flow (V):
Head loss is proportional to V^2 .
- iii. Pipe diameter (D):
Smaller diameter of the pipe results in larger head loss.
- iv. Roughness of pipe:
Rougher pipes create more resistance which increases the friction head loss.
- v. Type of flow:
Turbulent flow causes greater head loss than laminar flow.
- vi. Fluid properties:
Viscosity and density influence the friction factor.

4. Explain about the current hydropower generation and demand situation in Nepal. What are the reasons for slow development policy of Nepal?

Answer:

Nepal is the second richest country in terms of natural resources with more than 6,000 big and small rivers. Our country has enormous hydropower potential due to the steep mountainous topography and abundant fast flowing water resources. The theoretical hydropower potential of Nepal is approximately 83,000 MW out of which 42,000 MW hydropower have been identified as technically and economically feasible.

As of now, the total installed capacity of Nepal is 3878 MW and about 92.1% of the country's electricity generation is contributed by hydropower. During the monsoon season, the flow in rivers is high due to abundant rainfall which contributes to enough electricity generation for the self-use as well as for export to neighboring countries like India and Bangladesh. However, during dry season, electricity generation decreases significantly as most of the projects in Nepal are run-of-river schemes, resulting in lower river discharge and reduced electricity generation. The electricity demand in the country is also in increasing pattern due to urbanization, industrial growth, electrification of rural areas etc.

The slow development of Nepal's hydropower development policy is mainly due to:

- i. Political instability and frequent changes in government plans and priorities.
- ii. Poor infrastructure development that obstructs the smooth transportation of construction materials to the project site.
- iii. The rugged mountainous topography and complex geology of Nepal increase the risk as well as construction cost.
- iv. Delay in land acquisition, resettlement of affected communities, forest clearance, EIA requirements.

- v. Complex licensing and approval process, delay in obtaining permits from multiple governmental agencies.
- vi. Few storage-based projects which increases the seasonal variation in electricity.
- vii. Delay in grid expansion, insufficient transmission lines.

5. Explain about the current hydropower development policy of Nepal. Give your views for improvements needed in the policy?

Answer:

Nepal's hydropower sector is guided by the Hydropower Development Policy, 2058 (2001), along with subsequent acts, regulations, and energy sector reforms. The policy aims to develop the country's vast water resources for economic growth, energy security, and sustainable development. Some of the major objectives of the policy includes:

- i. To develop hydropower to meet the domestic electricity demand.
- ii. To promote the private sector participation in hydropower development.
- iii. To promote rural electrification and improve access to the electricity.
- iv. To ensure environmentally sustainable and socially responsible hydropower development.
- v. To encourage competition and transparency in project development and licensing.

The major features of the policy include:

- i. Small projects up to 1 MW do not require license which lowers barriers for micro and small hydropower.
- ii. Hydropower projects above 1 MW require license with a maximum validity of 50 years.
- iii. Encourages Independent Power Producers (IPPs).
- iv. Promotes Build-Own-Operate-Transfer (BOOT) and similar project models.
- v. Requires environmental Impact Assessment (EIA) or Initial Environmental Examination (IEE) as applicable.
- vi. The policy provides tax, royalty, customs, and foreign exchange incentives, especially for smaller projects and private developers.

The improvements needed in the Hydropower Policy are listed below:

- i. Storage hydropower projects must be prioritized and dependency on run-of-river plants should be reduced to ensure reliable electricity supply during dry seasons.
- ii. Complex licensing and approval process must be simplified and delays due to lengthy administrative processes must be eliminated.
- iii. Transmission line infrastructures must be strengthened by its simultaneous construction with generation process.
- iv. Domestic investment on the hydropower projects must be encouraged by expanding the financing opportunities through banks, bonds and public participation and reducing dependence on foreign loans.
- v. Fair compensation and rehabilitation must be ensured during the implementation of hydropower projects.
- vi. Environmental and social management should be adequately considered at every stage of construction.
- vii. Cross-border power trade must be enhanced by expanding the electricity export agreements.
- viii. Political stability must be given foremost priority, ensuring to maintain consistent policies and reduce frequent regulatory changes.
- ix. Digital monitoring, automation, smart grids must be promoted and operational efficiency must be improved.
- x. Climate change adaptation measures must be incorporated in hydropower planning and design.
- xi. Local communities must be ensured to be benefitted by employment, local infrastructure and share ownership.

6. Explain about the daily electricity demand variation in Nepal. Which types of hydropower project is needed to meet the gap in dry season? Explain with reasons?

Answer:

- Electricity demand in Nepal doesn't remain constant throughout the day due to changes in domestic, commercial, and industrial activities.
- The usage of electricity on the valley is usually at peak during mornings and evenings.
- The demand of electricity rises around 7 AM to 9 AM as people start to wake up and prepare for the day.
- During day time, the consumption of electricity is moderate from offices, shops and small factories.
- The highest load of the day occurs between 5 PM and 10 PM.
- Demand drops fast after 10 PM and stays low until early morning as the residential and industrial activities stops.
- This fluctuation of electricity demand during different time slots of the day is known as daily electricity variation.

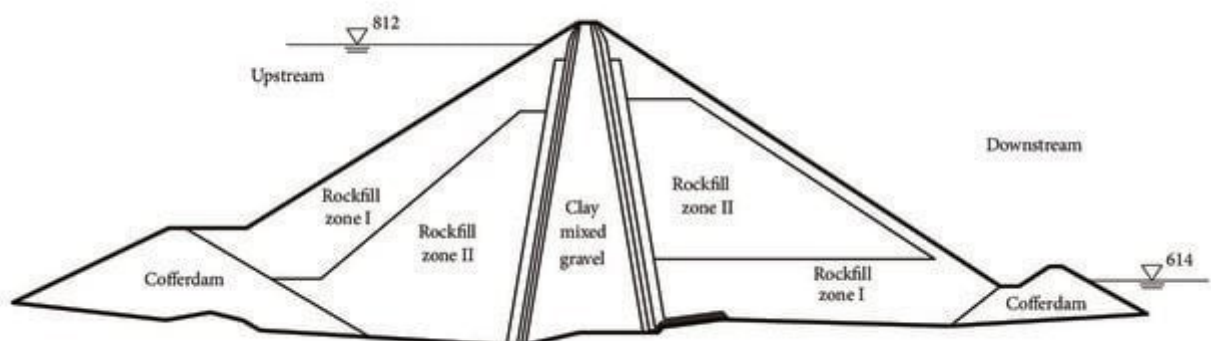
During the monsoon season, the flow in rivers is high due to abundant rainfall which contributes to enough electricity generation for the self-use as well as for export to neighboring countries like India and Bangladesh. However, during dry season, electricity generation decreases significantly as most of the projects in Nepal are run-of-river schemes, resulting in lower river discharge and reduced electricity generation. Hence, to meet the dry season gap, storage hydropower projects must be prioritized because of the following reasons:

- i. Storage hydropower projects stores excess water during the rainy season in a reservoir for the use in winter seasons when the river flow reduces.
- ii. It helps to maintain electricity production even when the river flows are low.
- iii. In these projects, water is released during morning and evening peak demand periods, which provides flexible operation and peak-load support.
- iv. It improves grid stability by responding quickly to the fluctuations in electricity demand.
- v. Storage projects also reduces the dependance on the imported electricity during the dry season.
- vi. It supports the integration of renewable energy sources such as solar and wind by providing dispatchable generation.

Most of the hydropower projects in Nepal is ROR hydropower projects. At present scenario, Nepal currently has only one operational storage hydropower project of Nepal i.e., the Kulekhani cluster (Kulekhani I, II, and III) in Makwanpur district. Some of the planned storage hydropower projects in Nepal are Budhi Gandaki storage hydropower plant, Tanahun storage hydropower plant and so on.

7. **Draw cross section of a typical high (say 300 m) Earth-Rock fill Dam, explain how the stability of its slope is determined.**

Answer:
earthe



The main components of the Earth-rock fill dam are listed below:

- i. Crest:
It is the top most part of the dam used for inspection and access.
- ii. Impervious clay core:
It prevents seepage through the dam.
- iii. Upstream rockfill shell:
It provides structural support to the dam.

- iv. Downstream rockfill shell:
It provides stability against sliding.
- v. Filter and transition zones:
This zone prevents migration of fine particles and safely drain the seepage.
- vi. Riprap:
It protects the upstream slope from wave action.
- vii. Toe drain and rock toe:
It collects seepage water and improve downstream stability, prevents the failure of dam by piping.
- viii. Foundation:
It is the natural ground supporting the dam.

The stability of the upstream and downstream slopes is evaluated by comparing resisting forces with the driving forces.

Factor of Safety:

The stability is measured by the Factor of Safety (FoS).

$$\text{FoS} = \frac{\text{Resisting Forces (Shear Strength)}}{\text{Driving Forces (Shear Stress)}}$$

Where :

Resisting forces = Cohesion $\wedge$ friction of soil / rock .

Driving forces = Self - weight of embankment

water pressure

seismic forces acting downslope .

If $\text{FoS} > 1.5$, then the dam is said to be stable .

If $\text{FoS} \approx 1.0$, then is said to be $\in$ Critical $\vee$ limiting equilibrium

If $\text{FoS} < 1.0$, then the dam becomes unstable .

8. What is hydraulic tunnel? Explain the condition of application and location of de-sanding basin in a hydropower project.

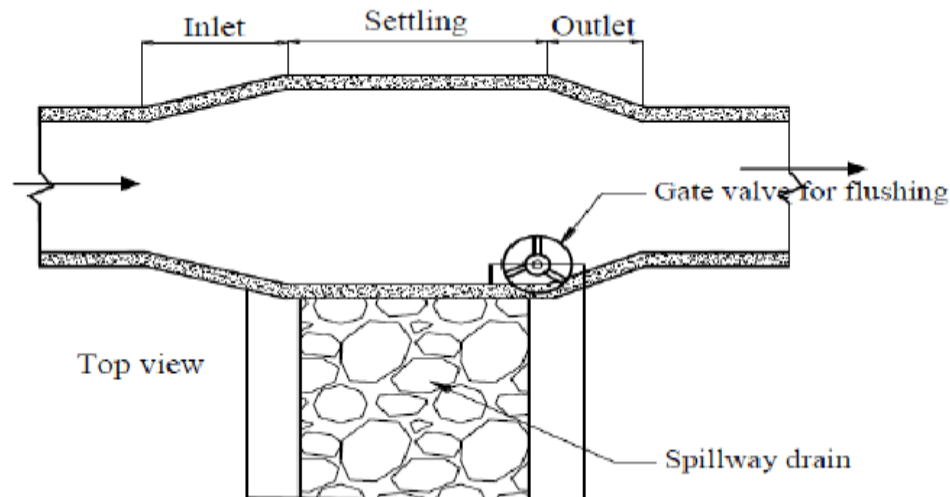
Answer:

Hydraulic tunnel is the water conduit formed by excavating mountain or hill without removing the overlaying rock. Hydraulic tunnel is provided under the following conditions:

- i. When the water passages have to be carried deep underground rather than in one open cut for the reasons of economics and safety.
- ii. When the open water passage is inevitably subjected to damages such as landslides, rockfall etc.
- iii. Where the water passage is carried across the densely populated areas.

Many construction projects in Nepal utilized hydraulic tunnel. Example: Upper Tamakoshi Hydropower project, Sunkoshi Marine Diversion Multipurpose project, Bheri Babai Diversion Multipurpose project, Melamchi water supply project and many more.

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.



The conditions for the application of de-sanding basin are as follows:

- i. **High Sediment Load:**
Rivers carrying large quantities of sand and silt, especially Himalayan rivers, require de-sanding basins.
- ii. **Run-of-River Hydropower Projects:**
Most run-of-river projects divert water directly from rivers without large reservoirs, making sediment removal essential.
- iii. **Protection of Hydraulic Equipment:**
To prevent excessive wear of turbines, valves, penstocks, and other hydraulic components.
- iv. **Coarse Sediment Presence:**
When sediment particles larger than the permissible size (typically 0.2 mm or larger) are present.
- v. **Long Headrace Canal or Tunnel:**
Sediment deposition in canals or tunnels can reduce carrying capacity and increase maintenance costs.
- vi. **Glacial or Mountain Rivers:**
Rivers originating from glaciers and steep mountain catchments carry heavy sediment, especially during the monsoon.

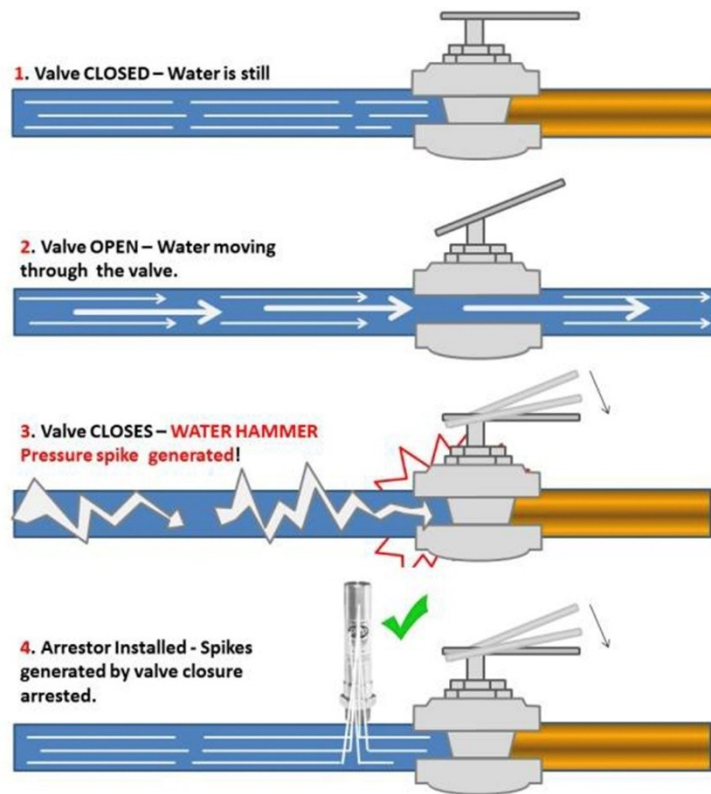
Location of de-sanding basin: The de-sanding basin is generally located,

- Immediately downstream of the intake (headworks).
- Before the headrace canal or headrace tunnel.
- Upstream of the forebay and penstock.
- On stable ground with sufficient space for construction and maintenance.
- At a location where flow can be slowed sufficiently to allow sediment to settle.

9. What do you understand by water hammer? Explain the several types of surge tank with schematic drawings.

Answer:

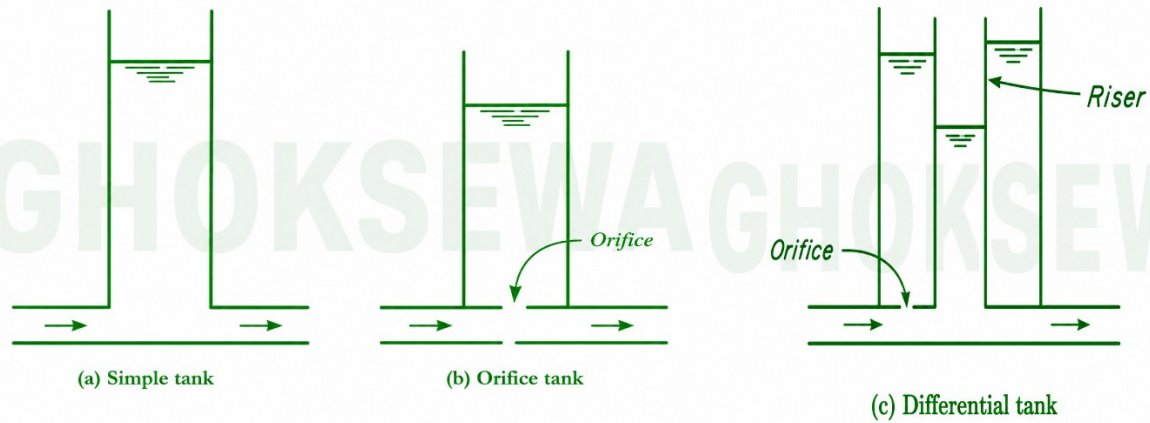
If a liquid flowing in a pipe is suddenly brought to rest by closing the valve, there is an abrupt rise in the pressure because the momentum of liquid is destroyed. A pressure wave is transmitted along the pipe. The velocity of this pressure wave is equal to the velocity of sound in the water. The pressure wave and the associated pulsation or fluctuations in the pressure produce the noise known as water hammer or knocking.



Surge tanks are provided to reduce the water hammer or elastic shock wave coming from the pressure pipeline or tunnel on closing and opening action of the valve.

Surge tank is the cylindrical storage tank which is connected to the penstock at suitable point. It is provided in the case of hydropower projects having long penstock pipes. Surge tank is divided into:

- Simple surge tank:**
 - It consists of a shaft connected to pressure tunnel directly or by a short connection of cross-sectional area not less than the area of the headrace tunnel.
 - It acts as a reservoir directly connected to the penstock line in such a manner that the water flows into and out of the tank without any head loss.
- Restricted orifice or throttled surge tank:**
 - It is a surge tank in which inlet is throttled to improve damping of oscillation by offering greater resistance and connected to the headrace tunnel with or without a connecting shaft.
 - It decreases the amplitude of the surge and effects more rapid reduction if the mass oscillation and gives economy up to 40% in comparison to the simple tank.
- Differential surge tank:**
 - It consists of a combination of the simple surge tank and the restricted orifice surge tank except that an internal riser pipe extends upward from the restricted orifice to near the top of the outside shell.
 - It separates out the two functions of simple storing of additional water and effective accelerating and decelerating of water in the main conduit by outside main tank and central riser respectively.



10. **Describe the selection criteria for different types of turbines used in hydroelectric plants.**

Answer:

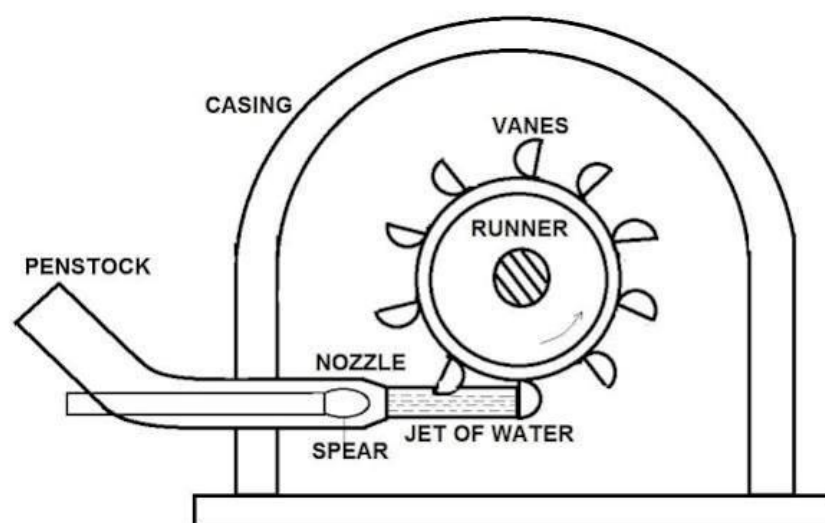
Turbine is the hydro-mechanical equipment that converts hydraulic energy into mechanical energy. It consists of wheel called runner provided with the number of curved vanes on the periphery. And guiding apparatus to direct the flow of water at the inlet in specified direction. The selection of a hydraulic turbine depends on the available head, discharge, power output, efficiency, operating conditions, and economic considerations. Selection of turbine is done such that it gives maximum efficiency, reliable operation, and minimum maintenance cost.

The different types of turbines used in hydroelectric plants are as follows:

i. **On the basis of type of energy at inlet:**

a. Impulse turbine:

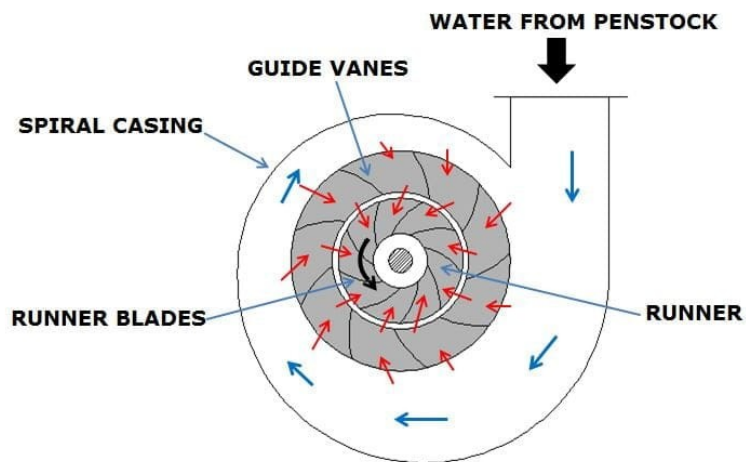
- The conversion of energy is due to change in momentum of water or due to the change in kinetic energy only is called impulse turbine.
- Water enters and leaves the turbine at atmospheric pressure in impulse turbine.
- It is suitable for high head and low discharges.
- Buckets are in action only when they are in front of the nozzle.
- Example: Pelton turbine



b. Reaction turbine:

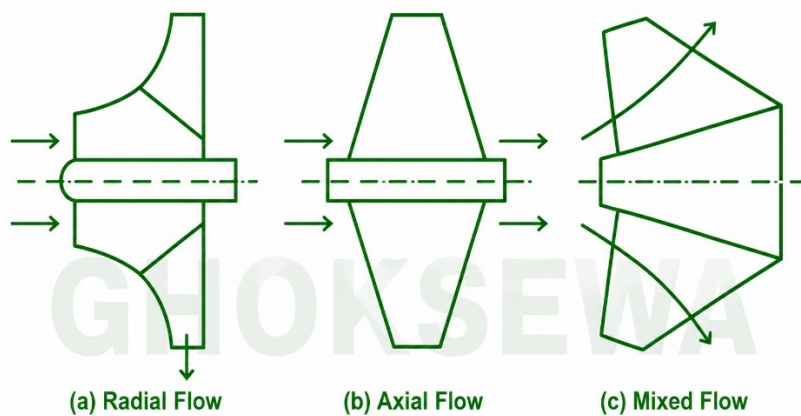
- The conversion of energy is due to the pressure difference at the two sides of vanes or due to the change in pressure of water.
- Water enters the turbine at high pressure and leaves the turbine at low pressure.

- It is suitable for low to medium head and medium to high discharges.
- The vanes are in action all of the time in reaction turbine.
- Example: Francis turbine, Kaplan turbine etc.



ii. On the basis of flow directions:

- Tangential flow turbine:**
 - The direction of flow is tangential to the shaft in tangential flow turbine.
 - Example: Pelton turbine
- Axial flow turbine:**
 - The direction of flow is parallel to the shaft in axial flow turbine.
 - Example: Kaplan turbine
- Radial flow turbine:**
 - The direction of flow is radial in radial flow turbine.
 - Example:
- Mixed flow turbine:**
 - The entry of flow is radial and exit is axial in mixed flow turbine.
 - Example: Francis turbine



iii. On the basis of specific speed:

- Low specific speed turbine:**

- The turbine whose specific speed is less than 50 rpm is known as low specific speed turbine.
- Example: Pelton turbine



b. Medium specific speed turbine:

- The turbine whose specific speed is equal to 50 to 260 rpm, is known as medium specific speed turbine.
- Example: Francis turbine



c. High specific speed turbine:

- The turbine whose specific speed is greater than 260 rpm, then it is known as high specific speed turbine.
- Example: Kaplan turbine

